AI/ML Algorithms, Dimensionality and Metrics

ML method: pretrained transformer encoder all-MiniLM-L6-v2 produces semantic vectors for both retrieved chunks and questions. Retrieval: ChromaDB vector similarity ranks candidate evidence. Generative model: Gemini synthesizes a grounded-language answer; MockLLM substitutes deterministic text in tests. Domain logic: structured AUTOSAR extraction and citation ID verification are deterministic rule-based components.

Dimensionality reduction: no separately fitted PCA, t-SNE or UMAP. Learning an embedding is not equivalent to applying PCA.

Threshold: normal default top_k=5, relevance threshold None. Top-k controls result count; a cutoff filters weak matches. Similarity scores are not probabilities of truth. Evaluator has a 0.3 default argument under a distinct evaluation path; do not confuse this with application default.

Evaluation available in source: retrieval success, citation ID validity, implementation-defined groundedness rate, negative-grounding accuracy and expected-keyword overlap. They are limited: a correctly cited generic answer can be useless or unsupported. Manual comparisons with the actual PDF are essential.

Why AI matters: semantic retrieval beyond exact keywords and engineering-text generation based on local documentary evidence; source provenance and testing are software engineering contributions. No newly trained model claimed.